

HOMEPLOTSAPPSLIVE EDITORINSERTVIEW

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Workspace

Name	Value
LTE_M	[0.05577,0.8]
NB_IoT	[0.0.0232,0.0]
SNR	[0.5,10,15,20]

5g and 6g exp 7.1.mlx

untitled * x

```
11 % Simulated Throughput (Mbps)
12
13 NB_IoT = 0.85*(1-exp(-SNR/8));
14
15 LTE_M = 1.2*(1-exp(-SNR/8));
16 disp('SNR (dB) NB-IoT(Mbps) LTE-M(Mbps)')
17 disp([SNR' NB_IoT' LTE_M'])
18 figure
19 % ----- Graph 1 -----
20 subplot(2,1,1)
21 plot(SNR,NB_IoT,'-o','LineWidth',2)
22 title('NB-IoT Throughput vs SNR')
23 xlabel('Signal-to-Noise Ratio (dB)')
24 ylabel('Throughput (Mbps)')
25 grid on
26 % ----- Graph 2 -----
27 subplot(2,1,2)
28 plot(SNR,LTE_M,'-s','LineWidth',2)
29 title('LTE-M Throughput vs SNR')
30 xlabel('Signal-to-Noise Ratio (dB)')
31 ylabel('Throughput (Mbps)')
32 grid on
```

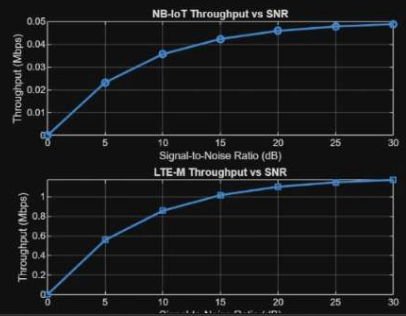
Command Window

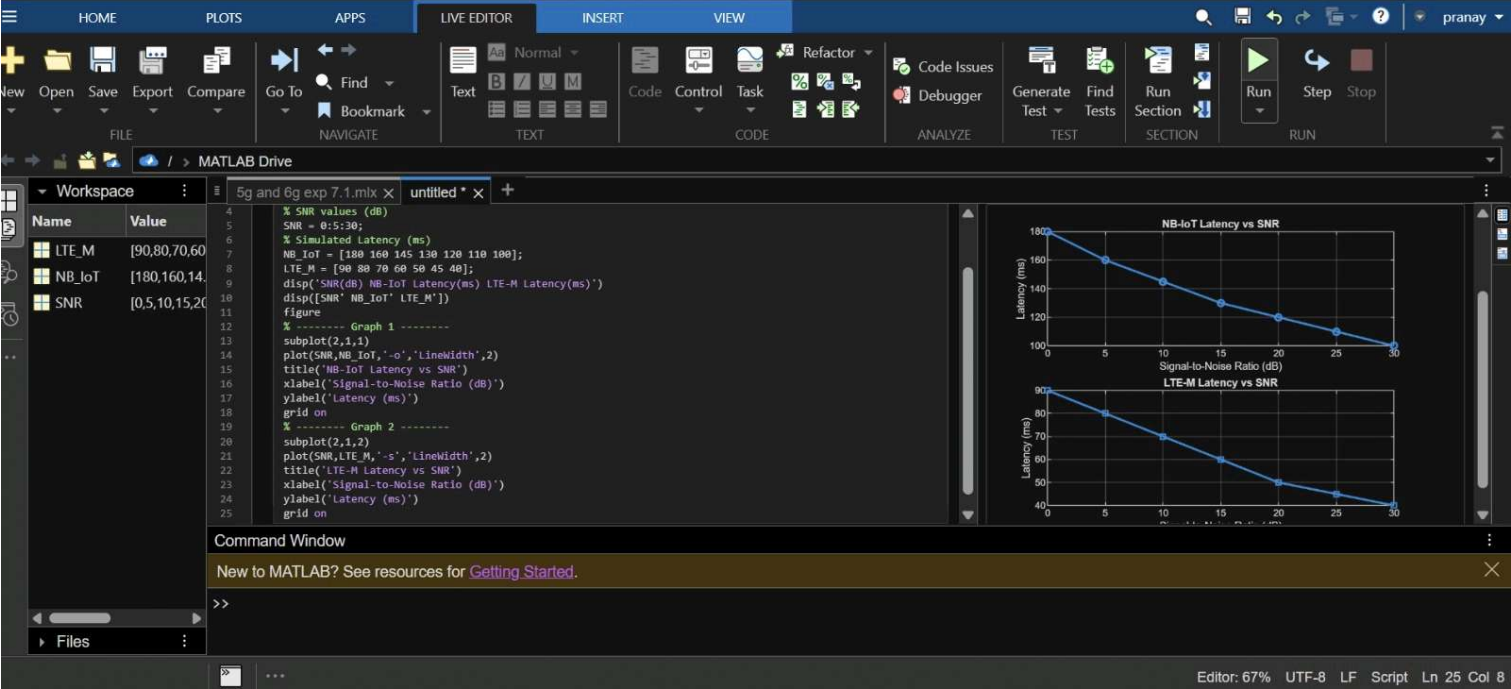
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Editor: 67% UTF-8 LF Script Ln 32 Col 8





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TextNormalText

CodeControlTaskRefactorCode IssuesDebuggerGenerate TestFind TestsRun SectionRunStepStop

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/ > MATLAB Drive

Workspace

Name	Value
LTE_M	[88,91,94,96]
NB_IoT	[82,86,89,92]
SNR	[0,5,10,15,20,25,30]

5g and 6g exp 7.1.mlx X

untitled X

```
4 % SNR values (dB)
5 SNR = 0:5:30;
6 % Simulated Packet Delivery Ratio (%)
7 NB_IoT = [82 86 89 92 95 97 99];
8 LTE_M = [88 91 94 96 98 99 100];
9 disp(['SNR' NB_IoT' LTE-M PDR(%)'])
10 disp(['SNR' LTE_M' NB-IoT PDR(%)'])
11 figure
12 % ----- Graph 1 -----
13 subplot(2,1,1)
14 plot(SNR,NB_IoT,'-o','LineWidth',2)
15 title('NB-IoT Packet Delivery Ratio vs SNR')
16 xlabel('Signal-to-Noise Ratio (dB)')
17 ylabel('Packet Delivery Ratio (%)')
18 grid on
19 % ----- Graph 2 -----
20 subplot(2,1,2)
21 plot(SNR,LTE_M,'-s','LineWidth',2)
22 title('LTE-M Packet Delivery Ratio vs SNR')
23 xlabel('Signal-to-Noise Ratio (dB)')
24 ylabel('Packet Delivery Ratio (%)')
25 grid on
```

Command Window

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